Two endpoint anchors on nine points

Human Compression project

We give a finite boundary calculation for transitive permutation representations with four marked involutions. Its purpose is to supply the small case in a separate induction concerning the high-rank representations studied in [1]. The calculation below does not assume that the generated group is symmetric or alternating, and does not use a pre-existing list of string C-groups.

Let Ω be a set of nine points and let h_0, h_1, h_2, h_3 be nonidentity involutions of Ω. Assume the string relations $[h_i, h_j] = 1$ whenever $|i - j| > 1$. Write $H_i = \langle h_j : j \neq i \rangle$. A transposition of h_i is called *eligible* if its two points belong to different H_i-orbits. An initial anchor is a point moved by h_0 and fixed by the other three marks; a final anchor is moved by h_3 and fixed by the other marks.

Theorem 1. *Suppose that $\langle h_0, h_1, h_2, h_3 \rangle$ is transitive, has distinct initial and final anchors, and that every mark has at least two eligible transpositions. Then a relabelling, sending the two anchors to 1 and 9, respectively, puts the ordered tuple in the form*

$$\begin{aligned} h_0 &= (1\ 2)(3\ 4), \qquad & h_1 &= (2\ 3)(4\ 5), \\ h_2 &= (5\ 6)(7\ 8), \qquad & h_3 &= (6\ 7)(8\ 9). \end{aligned} \tag{1}$$

This tuple generates the natural Alt(9) *and satisfies the full intersection property. For $H = \langle h_1, h_2, h_3 \rangle$ its orbits are*

$$\{1\}, \quad \{2, 3\}, \quad \{4, 5, 6, 7, 8, 9\}, \tag{2}$$

and the subgroup of $C_{\mathrm{Sym}(9)}(H)$ fixing both anchors is precisely $\langle (2\ 3) \rangle$.

Reduction to eleven endpoint pairs. Put $a = h_0$ and $d = h_3$. Let the anchors be b, c. Because a and d commute, $\{b, b^a\}$ is an orbit on which d acts trivially, and $\{c, c^d\}$ is an orbit on which a acts trivially. These orbits are disjoint: otherwise the two orbits coincide, whereas each generator acts nontrivially on one and trivially on the other. Relabel them as $\{1, 2\}$ and $\{8, 9\}$, with $b = 1, c = 9$. The commuting involutions a, d generate a Klein four-group. Each orbit on the remaining five points has size one, two, or four.

There are three labelled kinds of orbit of size two: only a moves its points, only d moves its points, or a and d induce the same swap. On an orbit of size four the action is regular and is unique up to relabelling. Let r_a, r_d, r_b, r_4, f be the numbers of these three kinds, regular orbits, and fixed points. They satisfy

$$2(r_a + r_d + r_b) + 4r_4 + f = 5. \tag{3}$$

Conversely, every nonnegative solution gives exactly one pair up to conjugacy fixing $1, 2, 8, 9$. Indeed one may match orbits of each labelled kind; on a regular orbit a chosen base point determines an equivariant bijection. Equation (3) has the eleven solutions in the table below. For each solution our representative assigns consecutive labels from $3, \ldots, 7$ first to a-only pairs, then d-only pairs, then common pairs, and finally regular orbits. On four consecutive labels x, y, z, w the last kind uses $a = (x\ y)(z\ w)$, $d = (x\ z)(y\ w)$. □

It remains to choose h_1, h_2. Both fix $1, 9$. There are exactly $7!/(2^j j!(7 - 2j)!)$ involutions with j swaps on the other seven points; summing over $j = 1, 2, 3$ gives 231. For each endpoint

representative, let B be those commuting with d and C those commuting with a. We examine every pair in $B \times C$. This supplies all the remaining string relations. Repeated marks can be rejected: if two marks coincide, omitting one leaves the other, so that the omitted mark has no eligible edge.

r_a	r_d	r_b	r_4	f	$\lvert B\rvert$	$\lvert C\rvert$	transitive	eligible
0	0	0	0	5	75	75	120	0
0	0	0	1	1	11	11	0	0
0	0	1	0	3	19	19	0	0
0	0	2	0	1	11	11	0	0
0	1	0	0	3	19	75	48	0
0	1	1	0	1	11	19	8	0
0	2	0	0	1	11	75	80	0
1	0	0	0	3	75	19	48	0
1	0	1	0	1	19	11	8	0
1	1	0	0	1	19	19	16	4
2	0	0	0	1	75	11	80	0

Here the last column imposes at least two eligible edges for every colour, after transitivity. In all, there are 11 507 pairs, 408 transitive tuples, and four survivors. These figures are complete finite calculations, with the following exact certificates and checks.

(i) `enumerate_boundary.py` generates all permutations of the seven unanchored points and keeps precisely the nonidentity involutions. For each of the eleven pairs it tests the two commutation conditions entry by entry. The connected components of the graph of all marked swaps are exactly the generated group orbits: each generator preserves these components and each edge is itself a generator step. The same procedure after omitting each colour gives exactly its omission orbits. It therefore computes transitivity and eligibility without a group library or a presupposed generated group.

(ii) The output gives all four literal surviving tuples and a bijection from each to (1). Each bijection is constructed from $1 \mapsto 1$ by propagating all four labelled edges. The certificate checks that every point gets one value, that the values are distinct, and that every marked edge is preserved. Both anchors have their specified images. Thus it proves ordered conjugacy, not merely equality of unlabelled graphs or of group orders.

(iii) `endpoint_control.py` separately enumerates matchings on six points. After fixing the two distinguished swaps it obtains 76 choices for each endpoint and 196 commuting pairs. The union of the conjugates of our eleven representatives by all permutations of $\{3, \dots, 7\}$ is exactly this literal set, with distinct endpoint types disjoint.

(iv) `gap_control.g` independently obtains the 231 involutions from the actual pointwise stabilizer of $1, 9$ in Sym(9). It uses actual subgroup orbits to repeat the full calculation. Its complete survivor set agrees with the Python list. For (1), it then checks equality of the generated subgroup with the natural Alt(9), and all $16^2 = 256$ equalities

$$\langle h_i : i \in I \rangle \cap \langle h_i : i \in J \rangle = \langle h_i : i \in I \cap J \rangle \qquad (I, J \subseteq \{0, 1, 2, 3\}).$$

Empty marked subsets are represented by the trivial subgroup. The group has order 181 440, but group equality and intersections are tested literally; the order alone is not used as an identification.

Completion of the proof. The orbit reduction and the exhaustive middle-pair calculation prove the classification in Theorem 1. The checks in (iv) prove the additional generation and intersection assertions. The omission graph for h_1, h_2, h_3 in (1) immediately gives (2). Any permutation centralizing H maps H-orbits to H-orbits of the same size, so it preserves each of these three orbits. On a transitive orbit an equivariant permutation fixing a point is the identity: if x is fixed, then $(x^h)^z = (x^z)^h = x^h$ for every $h \in H$. Consequently a centralizing permutation fixing 9 fixes the whole six-point orbit. It also fixes 1, and can only swap $2, 3$. That swap commutes with each of the three tail marks, so both possibilities occur. This proves the centralizer assertion. □

In the notation used for an outer extension, take $z = 1, p = 2, q = 3, w = 4$. The first mark is literally $(z\ p)(q\ w)$ and the only nonidentity anchored tail centralizer element is $(p\ q)$. No assertion about the outer extension, about higher degrees, or about the whole source manuscript is included here. The finite arithmetic is part of this boundary proof; no earlier small-degree classification is retained as a premise.

References

[1] P. J. Cameron, M. E. Fernandes and D. Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. This is the context for the separate induction; its diagrams and tables are not completeness premises here.